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Artificial Intelligence, Intelligent Systems, and Emerging Engineering Technologies: Recent Advances and Future Perspectives

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ABSTRACT

Artificial Intelligence (AI) has become one of the most impactful technologies of the modern world and has seen a lot of breakthroughs in the fields of engineering, healthcare, manufacturing, transportation, education, agriculture, finance, and smart cities. From machine learning, deep learning, reinforcement learning, to generative AI and edge intelligence, intelligent systems are rapidly advancing, allowing machines to tackle complex cognitive tasks with increasing proficiency and independence. Meanwhile, new engineering enabling technologies like the Internet of Things (IoT), Digital Twins, robotics and autonomous systems, blockchain, and cloud-edge computing are providing for highly connected, intelligent environments that enhance efficiency, reliability, and decision-making. The paper provides a detailed overview of the recent developments in the field of artificial intelligence and intelligent systems, focusing on their applications with the latest engineering technologies and how they are contributing to the real-world applications. It explores the advancements in foundation models, explainable AI, federated learning, multi-modality intelligence and AI-driven automation. Additionally, it explores several engineering applications such as smart healthcare, Industry 5.0, precision agriculture, intelligent transportation, cybersecurity, energy management, and environmental monitoring. While significant strides have been made, there are also some obstacles to overcome such as issues with data privacy, cyber security risks, model transparency, computational complexity, ethical concerns and regulatory compliance. The paper also outlines future research directions in the area of trustworthy AI, human-centric intelligent systems, sustainable AI, quantum-enhanced intelligence, autonomous engineering systems and collaborative human–AI environments. The review presented in this paper

gives researchers, engineers and practitioners a complete picture of the recent developments, challenges and future prospects within the fast-growing area of artificial intelligence and intelligent engineering technologies.

1. Introduction

Artificial Intelligence (AI) is one of the most transformative technologies of the twenty-first century, transforming the relationship between humans and machines and digital systems (Russell & Norvig, 2021; Wooldridge, 2021). AI allows computers to learn, reason, solve problems, make decisions, understand language, and visualize, which are tasks traditionally associated with human intelligence (Russell & Norvig, 2021; Minsky, 1986). The rapid progress of AI has been driven by increased computational power, the availability of large-scale datasets, and the development of sophisticated machine learning algorithms, making AI a pivotal technology in modern engineering, healthcare, transportation, finance, manufacturing, agriculture, and smart city applications (Mitchell, 1997; Bishop, 2006; Chollet, 2021). Intelligent systems utilize AI together with sensing, communication, computing, and automation technologies to understand their surrounding environment, analyze data, and make decisions with minimal human intervention (Russell & Norvig, 2021; Bishop, 2006). Such systems are increasingly integrated with emerging technologies such as the Internet of Things (IoT), cloud computing, edge computing, Digital Twins, robotics, blockchain, and advanced wireless communication networks (Gubbi et al., 2013; Xu et al., 2014; Grieves & Vickers, 2017; Tao et al., 2019). These technologies work together to facilitate real-time data collection, predictive analytics, intelligent control, and automated decision-making, thereby improving efficiency, reliability, and productivity across diverse application domains (Xu et al., 2014; Tao et al., 2019). In recent years, AI methodologies have achieved remarkable progress. Traditional machine learning techniques have evolved into powerful deep learning frameworks capable of processing massive volumes of structured and unstructured data (Mitchell, 1997; Goodfellow et al., 2016; LeCun et al., 2015). Significant breakthroughs have been achieved in computer vision, natural language processing, speech recognition, and multimodal learning through the development of Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers, and Vision Transformers (ViTs) (Krizhevsky et al., 2017; He et al., 2016; Vaswani et al., 2017; Dosovitskiy et al., 2021; Tan & Le, 2019). Moreover, the emergence of Generative Artificial Intelligence (Generative AI) and Large Language Models (LLMs) has significantly expanded AI capabilities in content generation, software development, scientific research, and intelligent decision-making (Brown et al., 2020; Devlin et al., 2019; OpenAI, 2023; Goodfellow et al., 2020).

AI has also become deeply integrated with modern engineering technologies, accelerating digital transformation across industries. The Internet of Things (IoT) continuously generates real-time data, while cloud and edge computing platforms provide scalable computational resources for intelligent data processing (Gubbi et al., 2013; Xu et al., 2014). A Digital Twin serves as a virtual representation of a physical system for monitoring, prediction, optimization, and lifecycle management (Grieves & Vickers, 2017; Tao et al., 2019). Likewise, Industry 5.0 promotes a new paradigm of human–machine collaboration, enabling adaptive, efficient, and sustainable manufacturing environments supported by intelligent automation and AI-driven analytics (Tao et al., 2019). These technological advances are

transforming engineering design, operational management, and industrial automation through data-driven intelligence.

AI has also demonstrated tremendous potential in scientific research and engineering innovation. In healthcare, AI supports disease diagnosis, medical image analysis, drug discovery, and remote patient monitoring (LeCun et al., 2015; Goodfellow et al., 2016). Intelligent transportation systems employ AI for autonomous driving, traffic management, and logistics optimization (Silver et al., 2016; Silver et al., 2017). In agriculture, AI enables precision farming, crop monitoring, and yield prediction, while financial institutions utilize AI for fraud detection, risk assessment, and intelligent customer services (Russell & Norvig, 2021; Bishop, 2006). Furthermore, AI-powered cybersecurity technologies play a critical role in threat detection and the protection of digital infrastructure, while high-performance AI benchmarking continues to improve deployment efficiency across real-world applications (Reddi et al., 2020). Despite these remarkable achievements, several challenges continue to hinder the widespread adoption of AI in critical applications. Issues related to data privacy, cybersecurity, algorithmic bias, limited model interpretability, and high computational requirements remain significant concerns (Russell & Norvig, 2021; Reddi et al., 2020). Many deep learning models operate as "black boxes," making their decisions difficult to interpret and reducing user trust, particularly in safety-critical domains such as healthcare and autonomous systems (Adadi & Berrada, 2018; Linardatos et al., 2021). To address these limitations, researchers are actively exploring Explainable Artificial Intelligence (XAI), Federated Learning, privacy-preserving AI, trustworthy AI, and energy-efficient intelligent computing (McMahan et al., 2017; Yang et al., 2019; Adadi & Berrada, 2018; Linardatos et al., 2021). These research directions aim to improve transparency, security, scalability, robustness, and sustainability while enabling intelligent systems that are reliable, trustworthy, and capable of supporting real-world decision-making. This paper presents a comprehensive review of recent advances in Artificial Intelligence, intelligent systems, and advanced machine learning technologies. It discusses the latest technological developments, key application domains, emerging engineering trends, current challenges, and future research directions that are expected to shape the next generation of intelligent engineering systems (OpenAI, 2023; Wooldridge, 2021; Chollet, 2021).

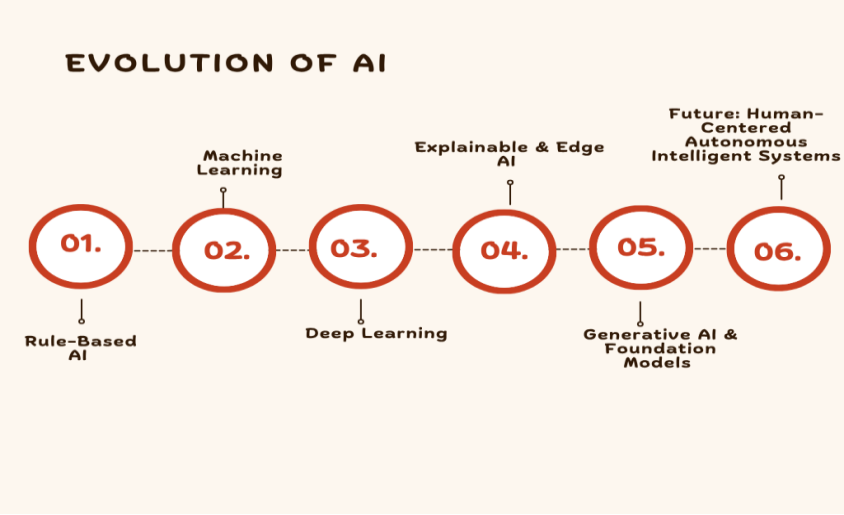


Figure 1. Evolution of Artificial Intelligence

Table 1. Summarize of Artificial Intelligence

Generation	Period	Major Technologies	Key Applications
Rule-Based AI	1950s–1980s	Expert Systems, Symbolic AI	Decision Support
Machine Learning	1990s–2010	SVM, Decision Trees, Random Forest	Pattern Recognition
Deep Learning	2012–2018	CNN, RNN, LSTM	Computer Vision, Speech Recognition
Explainable & Edge AI	2018–2022	XAI, Federated Learning, Edge AI	IoT, Smart Devices
Generative AI	2023–Present	Transformers, LLMs, Diffusion Models	Content Generation, Automation

2. Fundamentals of Artificial Intelligence and Intelligent Systems

Artificial Intelligence (AI) is the ability of computer systems to simulate human intelligence by emulating human learning, reasoning, problem-solving, decision making and language understanding (Russell & Norvig, 2021;). In traditional software, rules are already determined and written code performs the same functions each time; in contrast, AI systems are trained using data to learn patterns and continually improve their performance (Mitchell, 1997; Bishop 2006; Bengio, 2009). This capability has established AI as a key enabler for developing intelligent systems capable of operating efficiently in complex and dynamic environments (Gill et al., 2022; IEEE, 2025). According to Endsley (1995) and Russell and Norvig (2021), an intelligent system is a system designed to perform tasks with little human intervention that incorporates artificial intelligence, sensing technologies, data processing, machine learning, and intelligent decision making. The systems can gather and process environmental data, learn from past and current information, and provide intelligent decisions or recommendations (Endsley, 1995; Gill et al., 2022). Intelligent systems have been applied in various fields such as health care, robotics, industrial automation, autonomous vehicles, cybersecurity, smart agriculture and smart cities (Esteva et al., 2019; Topol, 2019; Schwab, 2017; Jeschke et al., 2017).

Machine Learning (ML) is one of the fundamental fields of AI where the computer learns from data without being directly programmed (Mitchell, 1997; Bishop, 2006). The ML algorithms are used to discover relationships in the data and apply them to prediction or classification (Mitchell, 1997; Bengio, 2009). MITCHELL (1997) and RUSSELL and NORVIG (2021) distinguish four main learning paradigms, namely supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning, which have various realworld applications. There are many advanced machine learning applications, such as image recognition, recommendation systems, predictive analytics, fraud detection and autonomous systems (Bishop, 2006; Esteva et al., 2019).

Deep Learning (DL) is an advanced branch of machine learning that uses multi-layer artificial neural networks to automatically learn multiple levels of features from raw data (Goodfellow et al., 2016;

LeCun et al., 2015; Bengio, 2009). Common deep learning architectures are Convolutional Neural Networks (CNNs) for image processing, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks for processing sequential data, and Transformer models for natural language understanding (Krizhevsky et al., 2017; He et al., 2016; Vaswani et al., 2017). Recently, Large Language Models (LLMs) have also revolutionized various applications like text generation, question answering, software development, and conversational AI (Brown et al., 2020; OpenAI, 2023; Yan et al., 2024). This is one of the most important research areas in the field of XAI: to enhance the interpretability and transparency of AI systems, leading to greater trust in users in safety-critical environments like healthcare, finance, and autonomous systems (Adadi & Berrada, 2018; Linardatos et al., 2021; Martínez-Fernández et al., 2021).

Edge AI is another significant development that would allow AI models to be run directly on edge devices, which would have a number of benefits in the field of the Internet of Things (IoT), wearable technology, and industrial monitoring systems, including reduced latency, enhanced privacy, and real-time intelligent decision-making (Xu et al., 2014; Jeschke et al., 2017; Gill et al., 2022). Generative Artificial Intelligence (Generative AI) has brought an array of revolutionary capabilities to the table, with the ability to generate new text, images, audio, video and software code (Goodfellow et al., 2020; OpenAI, 2023; Yan et al., 2024). In recent years, technologies such as Large Language Models (Kingma & Welling, 2014), Diffusion Models (Goodfellow et al., 2020), Variational Autoencoders (VAEs), and Generative Adversarial Networks (GANs) have revolutionized fields like engineering, healthcare, education, and creative industries (Yan et al., 2024).

Intelligent engineering systems (IESs) have become the result of the integration of AI with various other technologies including IoT, cloud computing, edge computing, Digital Twins, blockchain, and robotics (Gubbi et al., 2013; Xu et al., 2014; Grieves & Vickers, 2017; Tao et al., 2019; Jeschke et al., 2017). These technologies allow for real-time data collection, predictive analysis, automated monitoring and smart decision making across a wide field of engineering applications (Tao et al., 2019; Gill et al., 2022). In addition, new paradigms like Industry 4.0 and Industry 5.0 are driving the increasing need for AI-based automation while highlighting human-centric, resilient and sustainable industrial ecosystems (Schwab, 2017; European Commission, 2021). Other key scalable deployment and efficiency considerations for modern AI systems are federated learning, model compression and distributed computing (McMahan et al., 2017; Konecny et al., 2016; Han et al., 2016; Dean & Ghemawat, 2008).

To summarise, AI and intelligent systems are the cornerstone of the next generation of engineering, offering the potential for adaptive learning, intelligent automation, and data-driven decision-making (Gill et al., 2022; Gao & Adnan, 2025; Rahman et al., 2026). In the near future, the digital transformation is expected to mature further and new scientific, industrial, and societal AI solutions will be developed with the help of ongoing advancements in AI algorithms, computational infrastructure, intelligent networking, and new technologies like multi-agent systems and 6G communications (Wang et al., 2026; Tok et al., 2026; IEEE, 2025; IEEE, 2025).

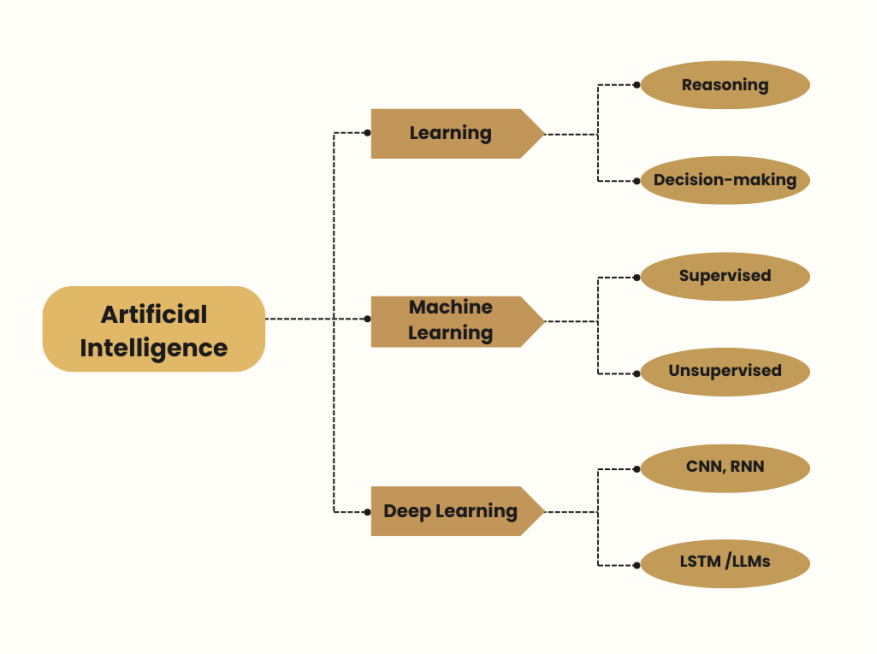


Figure 2. Hierarchical relationship among AI, Machine Learning, and Deep Learning.

Table 2. Comparison of Major AI Learning Paradigms

Learning Paradigm	Description	Representative Algorithms	Typical Applications
Supervised Learning	Learns from labeled data	SVM, Decision Tree, Random Forest, ANN	Disease diagnosis, Spam detection
Unsupervised Learning	Discovers hidden patterns in unlabeled data	K-Means, PCA, DBSCAN	Customer segmentation, Anomaly detection
Semi-Supervised Learning	Uses both labeled and unlabeled data	Self-training, Pseudo-labeling	Medical image analysis, Speech recognition
Reinforcement Learning	Learns through interaction using rewards	Q-Learning, Deep Q-Network (DQN), PPO	Robotics, Autonomous vehicles, Game AI
Deep Learning	Learns hierarchical features using neural networks	CNN, LSTM, Transformer, ViT	Image recognition, NLP, Healthcare

3. Recent Advances in Artificial Intelligence (2023–2026)

The evolution of Artificial Intelligence (AI) from 2023 to 2026 has been utterly transformative, marked by significant strides in machine learning, deep learning, foundation models, and high-performance

computing. The advances made in these areas have greatly enhanced the capabilities of intelligent systems to handle large-scale data, recognize complex patterns and assist in decision-making for a wide range of engineering and industrial applications. As resources such as cloud computing, specialized AI hardware and massive amounts of data have become more readily available, AI solutions are being deployed across a wide range of applications in healthcare, manufacturing, finance, transportation, education, and smart city development.

Few developments have had as much impact as the rise of Foundation Models and Large Language Models (LLMs). They are trained on vast amounts of data, and can be used for a wide range of tasks such as text generation, language translation, summarization, question answering, programming assistance, and scientific research. The main advantage of foundation models is that they offer a unified framework that can be fine-tuned for various applications.The key difference between traditional AI models and foundation models is that the latter can be adapted to different applications by fine-tuning. They have revolutionized human–computer interaction and enhanced research, educational, and software development productivity by understanding and producing human-like content. A significant leap forward is Generative Artificial Intelligence (GAI), which allows for the production of new text, images, videos, sounds and computer code. AI technologies such as transformer-based architectures, diffusion models, and generative adversarial networks (GANs) have prolonged the reach of AI into the world of engineering design, digital media, healthcare, and product development. Today, engineers have been harnessing generative AI to speed up software development, streamline product design, automate documentation, and foster innovation with intelligent content generation. Recently, there has been an increasing amount of research on Multimodal AI, which is the use of intelligent systems that combine information from several data sources, like text, images, audio, sensor signals and videos. This capability enhances the accuracy and reliability of AI models in various applications such as self-driving cars, medical diagnosis, intelligent surveillance, and human–computer interaction. Multi-modal systems can gain a deeper understanding of the real-world scenario and make more informed decisions by leveraging different modalities. In recent years, with a growing worry about transparency, fairness, and accountability, Explainable Artificial Intelligence (XAI) has grown in significance. Many deep-learning models are considered as black-box models where decisions are hard to understand. By generating explanations that are easy for people to understand, XAI techniques help build trust in AI predictions and pave the way for their use in industries like healthcare, finance, and industrial automation, where safety is paramount. One of the new trends is Edge AI and TinyML, which sees machine learning models being applied directly to resource-constrained devices like wearable sensors, smartphones, drones and IoT devices. These technologies lower communication latency, boost privacy, cut bandwidth and allow for real-time decision making without continuous cloud connectivity. Edge AI plays a crucial role in smart healthcare, predictive maintenance, environmental monitoring, and intelligent manufacturing. While significant progress has been made, there are ongoing challenges including data privacy, computational complexity, energy requirements, algorithmic bias, and ethical issues. Self-learning AI systems are anticipated to be more trustworthy, efficient, explainable, and human-centric and enable sustainable engineering solutions and autonomous intelligent systems.

Table 3. Recent AI Technologies and Their Applications

Technology	Key Features	Major Applications
Foundation Models	Multi-task learning	NLP, Research, Software Development
Generative AI	Content generation	Design, Healthcare, Education
Multimodal AI	Multi-source data fusion	Robotics, Autonomous Vehicles
Explainable AI	Transparent decision-making	Healthcare, Finance
Edge AI & TinyML	Real-time on-device intelligence	IoT, Wearables, Smart Manufacturing

4. Emerging Engineering Technologies

The new engineering technology is changing the face of traditional engineering work by combining the power of Artificial Intelligence (AI) with cutting-edge digital technology to build intelligent, autonomous and interconnected systems. Industries have been undergoing digital transformation at an unprecedented pace thanks to the marriage of Artificial Intelligence and Internet of Things (IoT), Digital Twins, robotics, cloud-edge computing, blockchain, autonomous systems and next-generation communication networks. The use of these technologies can facilitate real-time monitoring, predictive analytics, intelligent automation, and data-driven decision-making, resulting in improved operational efficiency, cost savings, and system reliability. Intelligent engineering is made possible by connecting physical devices through sensors, communication networks, and cloud platforms, which form the basis of the Internet of Things (IoT). AI algorithms can run on billions of IoT devices to analyze real-time data and identify any anomalies, predict equipment failures, optimize resource usage, and improve overall operational efficiency. AI-powered IoT systems are found in various applications, including smart homes, healthcare monitoring, industrial automation, environmental monitoring, and smart agriculture. The other new and rapidly emerging technology is the Digital Twin, which generates a virtual copy of a physical system, process or asset. A Digital Twin receives real-time data from sensors and constantly updates its virtual model to simulate the behavior of the system, monitor its performance and predict future behavior. Smart manufacturing, aerospace, healthcare, energy management and infrastructure monitoring are among the many areas leveraging AI-powered Digital Twins for predictive maintenance, fault diagnosis, and operational optimization. AI and Digital Twins can be used together to improve decision-making by simulating data and making intelligent recommendations in real-time.The combination of **cloud computing and edge computing** has further strengthened intelligent engineering systems. Cloud computing provides scalable storage and computational resources for training complex AI models, while edge computing processes data near the source, reducing latency and improving response time. This hybrid architecture enables real-time analytics in applications such as autonomous vehicles, industrial automation, healthcare monitoring, and smart city infrastructure. Processing sensitive data at the edge also improves privacy and reduces network bandwidth requirements.

Blockchain is another new technology that offers decentralized, secure and tamper-proof data management. Combined with AI, blockchain can provide added value in terms of data integrity, transparency, and trust in distributed engineering systems. The application of AI-blockchain integration has expanded to various sectors including supply chain management, healthcare data sharing, financial services, and industrial Internet of Things (IIoT) networks, where the need for secure data exchange and reliable decision-making is critical.

The technologies of 5G and 6G communication are also shaping future engineering systems, with 5G offering ultra-low latency, high-speed connectivity, and massive device communication. The cutting-edge communication infrastructure is enabling real-time AI applications, such as autonomous vehicles, telemedicine, industrial robotics, smart manufacturing, and intelligent city infrastructure. These new technologies are moving towards the creation of extremely connected, adaptive and autonomous engineering ecologies that look to tackle complex industrial and societal challenges. In conclusion, the synergy of AI and these new engineering technologies is transforming the landscape of modern engineering, offering ways to optimize the engineering process, improve operational efficiency, and create more sustainable, eco-friendly solutions. Ongoing research and innovation in these areas will be key to the development of next-generation engineering systems that will be more efficient, resilient and environmentally friendly.

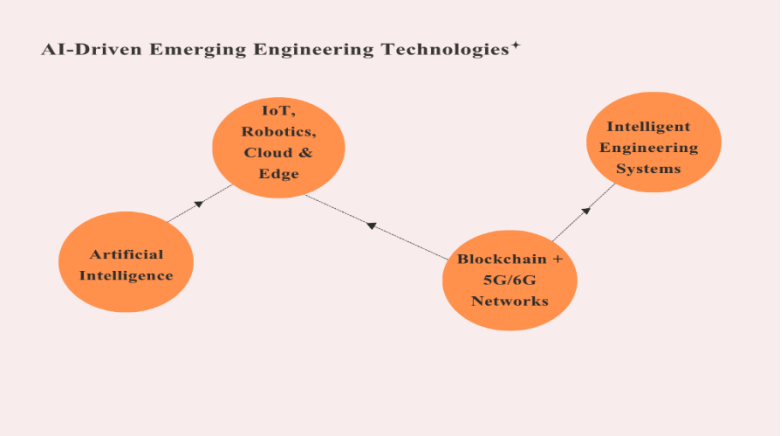


Figure 4. Integration of Artificial Intelligence with emerging engineering technologies.

Table 4. Emerging Engineering Technologies

Technology	Key Features	Major Applications
Internet of Things (IoT)	Real-time sensing and connectivity	Smart homes, agriculture, healthcare
Robotics	Intelligent automation	Industrial robots, autonomous drones
Cloud & Edge Computing	Scalable and low-latency computing	Smart cities, healthcare, Industry 5.0

Blockchain	Secure and decentralized data sharing	Supply chain, finance, IIoT
5G/6G Networks	High-speed, low-latency communication	Autonomous vehicles, remote healthcare

5. Applications of Artificial Intelligence and Intelligent Systems

Artificial Intelligence (AI) and intelligent systems have become an integral part of the modern development of engineering and industry, which allow not only automation, but also predictive analytics and intelligent decision making in various application areas. The applications of AI in high-sensing, high-communication technologies, and high-computational power have significantly advanced efficiency, accuracy, productivity, and lowered the cost of operations. The use of Artificial Intelligence (AI) solutions in the healthcare, manufacturing, transportation, agriculture, finance, education, cybersecurity and smart city infrastructure sectors today is revolutionizing the industry. AI is also one of the fundamental tools in the manufacturing industry for the creation of smart factories and Industry 5.0. Intelligent manufacturing systems employ machine learning, computer vision and Industrial Internet of Things (IIoT) technologies to monitor manufacturing processes, predict when equipment is failing, and optimize resource usage. Predictive maintenance models dig into sensor information to forecast potential machine failures beforehand, minimizing downtime and expenses in maintenance. The collaborative robots (cobots) also contribute to an increase in productivity by operating alongside the human operator in intricate manufacturing settings in a safe manner.

In agriculture, AI is used to assist in precision farming, which involves using data from drones, IoT sensors, and satellites to analyze the environment and crop conditions. Intelligent systems help farmers keep track of soil moisture, identify crop diseases, forecast weather conditions, manage irrigation schedules and estimate crop yields. The technologies boost agricultural productivity and reduce water, fertilizer and pesticide consumption, which helps to promote sustainable farming practices and food security globally.

AI has been extensively used in the financial industry for fraud detection, risk management, credit scoring, algorithmic trading, and customer support automation. Financial fraud is combated by ingesting vast amounts of financial transactions into machine learning algorithms which are used to detect suspicious activity. AI chatbots and virtual assistants offer customer support tailored to individual needs and predictive analytics help financial services organizations make informed investment and risk management decisions.

AI's role in threat detection, malware classification, intrusion detection, and network anomaly analysis is also a fast-expanding field, particularly in the realm of cybersecurity. Intelligent security systems are always on the lookout for real-time cyberattacks and network traffic. Using AI in security solutions provides better response times, increased system strength, and defence against changing cyber threats.

Additionally, smart cities combine AI with the Internet of Things, cloud computing and intelligent transportation systems, to manage urban infrastructure and services effectively. Smart city applications of AI range from traffic management to energy management, waste management, environmental monitoring, public safety and emergency response. These technologies enhance the quality of urban life, improve sustainability, efficient use of resources and environmental protection.

In conclusion, the integration of AI and intelligent systems across various industries is transforming the engineering landscape and society at large, driving operations, decision-making, and solutions to complex problems without human intervention. The potential of AI is still growing with its applications projected to be expanded in the future to encompass the next generation of intelligent engineering systems, digital transformation and sustainable development.

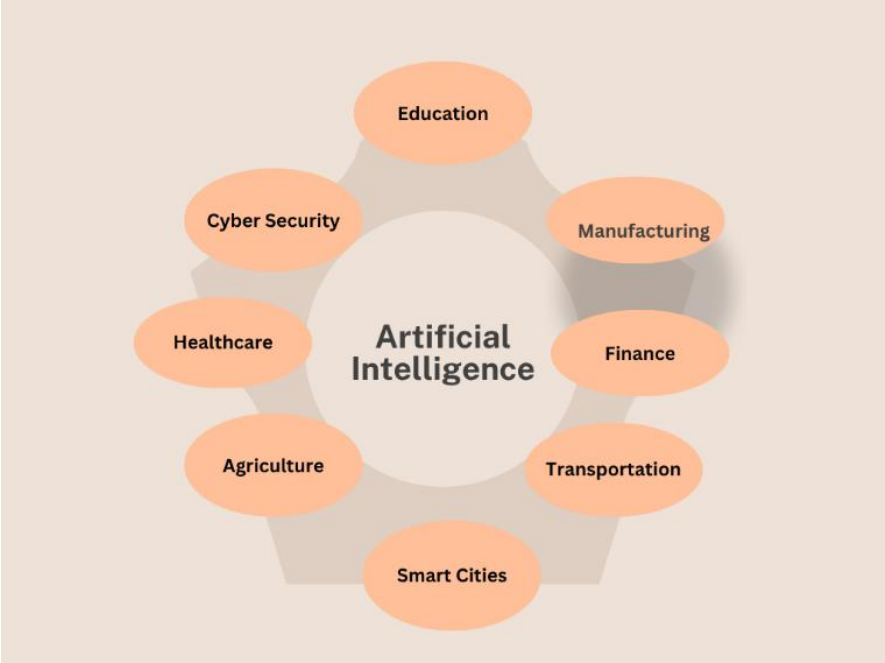


Figure 5. Applications of AI and intelligent systems.

Table 5. AI Applications Across Different Sectors

Sector	**AI Technologies**	**Major Benefits**
Healthcare	Deep Learning, Computer Vision	Disease diagnosis, remote monitoring
Manufacturing	Machine Learning, IIoT, Robotics	Predictive maintenance, quality control
Transportation	Computer Vision, Reinforcement Learning	Autonomous driving, traffic optimization
Agriculture	IoT, Drones, AI Analytics	Precision farming, crop monitoring
Finance	Machine Learning, NLP	Fraud detection, risk analysis

Education	Intelligent Tutoring Systems, Generative AI	Personalized learning, automated assessment
Cybersecurity	Deep Learning, Anomaly Detection	Threat detection, intrusion prevention
Smart Cities	AI, IoT, Edge Computing	Traffic management, energy optimization

6. Challenges, Ethical Issues, and Future Research Directions

Still, some limitations hinder the general adoption of AI and intelligent systems.However, there are some technical, ethical and societal issues that have hindered the widespread adoption of AI and intelligent systems. The critical applications of AI, including healthcare, transportation, manufacturing, and finance, are increasingly gaining traction in the industry, which has made the task of ensuring reliability, security, transparency, and fairness of these systems a top priority in the research field.

To create trustworthy and sustainable intelligent systems, it is necessary to address these challenges. Data quality and availability is one of the key challenges. To train AI models, you need a vast and rich set of high-quality data that's properly labeled. The acquisition and annotation of these types are, however, typically costly, time consuming and regulated by privacy concerns. In many engineering scenarios, missing, noisy or imbalanced data can dramatically affect model performance and generalizations. Thus, it is still an important research topic to design data-efficient learning approaches and powerful preprocessing methods.

The second main issue is the explainability of AI models. While the predictive accuracy of deep learning models is very high, it is often difficult to comprehend the decision-making process in such models due to their complex internal structure. The opaque nature is often referred to as the black box problem, which reduces trust by the users, particularly in safety-critical systems such as the autonomous driving system, medical diagnosis, or industrial control systems. Explainable Artificial Intelligence (XAI) has come to the forefront as a viable alternative, offering explanations in a manner that is comprehensible to humans to foster trust, accountability, and regulatory compliance.

Privacy and cybersecurity are another major part of AI-Powered Systems. Intelligent applications often handle personal and industrial data that is sensitive and is a target for cyberattacks and unauthorized access. Potential system failures, malicious attacks, and data manipulation can threaten system reliability and privacy. To ensure high predictive performance while bolstering the security of AI systems, privacy-preserving approaches like federated learning, differential privacy, secure multi-party computation, and blockchain-based data management are actively pursued.

Ethical issues are also a major part of the deployment of AI. The use of algorithmic bias, due to the imbalance or unrepresentation of training data, can create an unfair or discriminatory decision. Therefore, guaranteeing fair, accountable, transparent, and responsible governance of AI is crucial for developing socially acceptable intelligent systems. There are growing international norms and

approaches to encourage responsible development and application of AI technologies.International organizations and governments are increasingly creating ethical guidelines and regulatory frameworks to encourage responsible development and application of AI technologies. Future research is likely to address developing trustworthy, explainable, sustainable AI systems that will work efficiently in real-world contexts. Some research priorities are emerging such as Human centered AI – intelligent systems that work alongside humans, Green AI – minimizing computational complexity and energy usage, and Quantum AI – harnessing quantum computing and machine learning to solve highly complex optimization problems. Another category of technologies that will be vital to the next generation of engineering technologies is AI-driven Digital Twins, autonomous robotics, multimodal intelligence, edge AI, and intelligent systems with 6G connectivity. The progress will enable real-time decision making, predictive analytics and intelligent automation in various application areas. Overall, addressing current technical and ethical hurdles, and fostering responsible innovation, will be crucial to the future success of AI and intelligent systems. To develop secure, transparent, and human-centric AI solutions contributing to sustainable engineering and society, continued interdisciplinary research, academia–industry collaborations, and the design of robust regulatory frameworks will all be necessary.

Table 6. Challenges and Future Research Opportunities

Challenge	**Impact**	**Future Research Direction**
Data quality and availability	Reduced model accuracy	Data-efficient learning and synthetic data generation
Black-box AI models	Low transparency and trust	Explainable Artificial Intelligence (XAI)
Privacy and cybersecurity	Data leakage and cyber threats	Federated learning, blockchain, differential privacy
Algorithmic bias	Unfair and unreliable decisions	Fair and ethical AI frameworks
High computational cost	Increased energy consumption	Green AI and energy-efficient algorithms
Real-time deployment	Latency and scalability issues	Edge AI, TinyML, and 6G-enabled AI systems

7. Conclusion

The advent of Artificial Intelligence (AI), intelligent systems, and new engineering technologies is revolutionizing the way engineering problems are being solved in today's world, providing intelligent automation, predictive analytics and data-driven decision making. Intelligent systems have seen a dramatic increase in their capabilities in a wide range of application areas, due to the rapid advances in machine learning, deep learning, generative AI, the Internet of Things (IoT), robotics, edge

computing, and cloud technologies. These technologies have enhanced the efficiency, accuracy, reliability and sustainability measures in healthcare, manufacturing, transportation, agriculture, finance, cyber security, education, and smart city development. This review has given a comprehensive overview of the recent developments in AI and underlined the integration of intelligent systems with the new engineering technologies. The presentation showcased the application of AI-driven solutions in the context of real-time monitoring, predictive maintenance, autonomous decision-making, and intelligent resource management. The paper also explored some of the most significant technological advancements such as foundation models, generative AI, multimodal learning, explainable AI, federated learning, and edge intelligence that are driving the evolution of engineering systems. While significant strides have been made, there are still several challenges to overcome in order for AI to become widely adopted and reliable. Future work is required through research and interdisciplinary collaboration between academia, industry and policymakers to overcome problems of data quality, computational complexity, model interpretability, privacy protection, cyber security, algorithmic bias, and ethical governance. Safeguarding these challenges will be key to creating safe, transparent, reliable, and people-centric AI systems that can function in the real world.

Future research areas are anticipated to include Trustworthy AI, Green AI, Quantum machine learning, Autonomous Engineering Systems, Digital Twin technologies and Collaborative human–AI environments. The level of digital transformation in engineering will further accelerate with the integration of AI with next generation communication networks, intelligent sensing and sustainable computing platforms. With the ongoing advancements in AI technology, it is expected that it will continue to be a critical part of solving global challenges, optimizing industrial processes, enriching the quality of life, and supporting sustainable development. Hence, continuous innovation, responsible use of Artificial Intelligence and collaboration between various disciplines will play an important role in the years to come, when it comes to fully exploiting the potential of Artificial Intelligence and intelligent engineering technologies.

References

[1] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2021.
[2] Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
[3] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
[4] T. Mitchell, *Machine Learning*. McGraw-Hill, 1997.
[5] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
[6] Vaswani *et al.*, "Attention Is All You Need," in *Proc. NeurIPS*, 2017.
[7] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL-HLT*, 2019.
[8] T. B. Brown *et al.*, "Language Models are Few-Shot Learners," in *Proc. NeurIPS*, 2020.
[9] OpenAI, "GPT-4 Technical Report," 2023.
[10] D. Silver *et al.*, "Mastering the game of Go with deep neural networks and tree search," *Nature*, vol. 529, pp. 484–489, 2016.
[11] D. Silver *et al.*, "Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm," *arXiv:1712.01815*, 2017.
[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proc. CVPR*, 2016.

A. Krizhevsky, I. Sutskever, and G. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," *Communications of the ACM*, vol. 60, no. 6, pp. 84–90, 2017.

[13] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," *arXiv:1312.6114*, 2014.

[14] Goodfellow *et al.*, "Generative Adversarial Networks," *Communications of the ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[15] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked Autoencoders Are Scalable Vision Learners," in *Proc. CVPR*, 2022.

[16] M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in *Proc. ICML*, 2019.

[17] Dosovitskiy *et al.*, "An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale," in *Proc. ICLR*, 2021.

[18] M. Minsky, *The Society of Mind*. Simon & Schuster, 1986.

[19] F. Chollet, *Deep Learning with Python*, 2nd ed. Manning Publications, 2021.

[20] M. Wooldridge, *A Brief History of Artificial Intelligence*. Flatiron Books, 2021.

[21] M. Grieves and J. Vickers, "Digital Twin: Mitigating Unpredictable, Undesirable Emergent Behavior in Complex Systems," Springer, 2017.

[22] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital Twin in Industry: State-of-the-Art," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 4, pp. 2405–2415, 2019.

[23] L. Da Xu, W. He, and S. Li, "Internet of Things in Industries: A Survey," *IEEE Transactions on Industrial Informatics*, vol. 10, no. 4, pp. 2233–2243, 2014.

[24] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, "Internet of Things: A Vision, Architectural Elements and Future Directions," *Future Generation Computer Systems*, vol. 29, no. 7, pp. 1645–1660, 2013.

[25] B. McMahan *et al.*, "Communication-Efficient Learning of Deep Networks from Decentralized Data," in *Proc. AISTATS*, 2017.

[26] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated Machine Learning: Concept and Applications," *ACM Transactions on Intelligent Systems and Technology*, vol. 10, no. 2, 2019.

[27] Adadi and M. Berrada, "Peeking Inside the Black Box: A Survey on Explainable Artificial Intelligence (XAI)," *IEEE Access*, vol. 6, pp. 52138–52160, 2018.

[28] P. Linardatos, V. Papastefanopoulos, and S. Kotsiantis, "Explainable AI: A Review of Machine Learning Interpretability Methods," *Entropy*, vol. 23, no. 1, p. 18, 2021.

[29] V. J. Reddi *et al.*, "MLPerf Inference Benchmark," *IEEE Micro*, vol. 40, no. 2, pp. 50–58, 2020.

[30] S. Han, H. Mao, and W. Dally, "Deep Compression," *International Conference on Learning Representations*, 2016.

[31] J. Dean and S. Ghemawat, "MapReduce: Simplified Data Processing on Large Clusters," *Communications of the ACM*, vol. 51, no. 1, pp. 107–113, 2008.

[32] K. Schwab, *The Fourth Industrial Revolution*. Crown Publishing, 2017.

[33] European Commission, "Industry 5.0: Towards a Sustainable, Human-Centric and Resilient European Industry," 2021.

[34] M. R. Endsley, "Situation Awareness in Engineering Systems," *Human Factors*, vol. 37, no. 1, pp. 32–64, 1995.

[35] S. Jeschke, C. Brecher, H. Song, and D. B. Rawat, *Industrial Internet of Things*. Springer, 2017.

[36] J. Konecny, H. B. McMahan, and D. Ramage, "Federated Optimization," *arXiv:1610.02527*, 2016.

[37] Y. Bengio, "Deep Learning of Representations," *Foundations and Trends in Machine Learning*, vol. 2, no. 1, pp. 1–127, 2009.

[38] Esteva *et al.*, "A Guide to Deep Learning in Healthcare," *Nature Medicine*, vol. 25, pp. 24–29, 2019.

[39] E. Topol, *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*. Basic Books, 2019.

[40] S. S. Gill *et al.*, "AI for Next Generation Computing: Emerging Trends and Future Directions," *Internet of Things*, vol. 19, 2022.

[41] S. Martínez-Fernández *et al.*, "Software Engineering for AI-Based Systems: A Survey," *ACM Computing Surveys*, 2021.

[42] P. Gao and M. Adnan, "Overview of Emerging Electronics Technologies for Artificial Intelligence: A Review," *Materials Today Electronics*, 2025.

[43] "Artificial Intelligence (AI): Foundations, Trends and Future Directions," *Telematics and Informatics Reports*, vol. 20, 2025.

[44] X. Yan, Y. Xiao, and Y. Jin, "Generative Large Language Models Explained," *IEEE Computational Intelligence Magazine*, vol. 19, no. 4, 2024.

[45] Z. Wang *et al.*, "Multi-Agent Systems: From Classical Paradigms to Large Foundation Model-Enabled Futures," *IEEE/CAA Journal of Automatica Sinica*, vol. 13, no. 5, pp. 1007–1023, 2026.

[46] Rahman *et al.*, "Artificial Intelligence Innovations, Challenges and Emerging Trends in Engineering Education," *Discover Education*, vol. 5, 2026.

[47] Y. E. Tok *et al.*, "Artificial Intelligence for Next-Generation 6G Technologies and Networks," *Discover Networks*, vol. 2, 2026.

[48] IEEE, "Artificial Intelligence Research and Trends," IEEE AI Portal, 2025.

[49] IEEE, "Publications on Artificial Intelligence," IEEE AI Publications Portal, 2025.